It is hardly surprising that ants are exploited by parasites, not just other ants but an astonishing menagerie of specialist hangers-on. Worker ants sweep a rich flow of food from a wide catchment area into a central hoard which is a sitting target for freeloaders. Ants are also good agents of protection: they are well-armed and numerous. The aphids of Chapter 10 could be seen as paying out nectar to hire professional bodyguards. Several butterfly species live out their caterpillar stage inside an ants' nest. Some are straightforward pillagers. Others offer something to the ants in return for protection. Often they bristle, literally, with equipment for manipulating their protectors. The caterpillar of a butterfly called *Thisbe irenea* has a sound-producing organ in its head for summoning ants, and a pair of telescopic spouts near its rear end which exude seductive nectar. On its shoulders stands another pair of nozzles, which cast an altogether more subtle spell. Their secretion seems to be not food but a volatile potion that has a dramatic impact upon the ants' behaviour. An ant coming under the influence leaps clear into the air. Its jaws open wide and it turns aggressive, far more eager than usual to attack, bite, and sting any moving object. Except, significantly, the caterpillar responsible for drugging it. Moreover, an ant under the sway of a dope-peddling caterpillar eventually enters a state called 'binding', in which it becomes inseparable from its caterpillar for a period of many days. Like an aphid, then, the caterpillar employs ants as bodyguards, but it goes one better. Whereas aphids rely on the ants' normal aggression against predators, the caterpillar administers an aggression-arousing drug and it seems to slip them something addictively binding as well.

I have chosen extreme examples. But, in more modest ways, nature teems with animals and plants that manipulate others of the same or of different species. In all cases in which natural selection has favoured genes for manipulation, it is legitimate to speak of those same genes as having (extended phenotypic) effects on the body of the manipulated organism. It doesn't matter in which body a gene physically sits. The target of its manipulation may be the same body or a different one. Natural selection favours those genes that manipulate the world to ensure their own propagation. This leads to what I have called the central theorem of the extended phenotype: *An animal's behaviour tends to maximize the survival of the genes 'for' that behaviour, whether or not those genes happen to be in the body of the particular animal performing*